CLASS 10 · CBSE · NCERT MATHEMATICS · FORMULA BANK

Chapter 8: Introduction to Trigonometry

Every formula and result of the chapter on one sheet

1	Sine	$\sin A = \frac{\text{opposite}}{\text{hypotenuse}}$
2	Cosine	$\cos A = \frac{\text{adjacent}}{\text{hypotenuse}}$
3	Tangent	$\tan A = \frac{\text{opposite}}{\text{adjacent}} = \frac{\sin A}{\cos A}$
4	Cosecant	$\operatorname{cosec} A = \frac{1}{\sin A}$
5	Secant	$\sec A = \frac{1}{\cos A}$
6	Cotangent	$\cot A = \frac{1}{\tan A} = \frac{\cos A}{\sin A}$
7	Identity 1	$\sin^2 A + \cos^2 A = 1$
8	Identity 2	$1 + \tan^2 A = \sec^2 A$
9	Identity 3	$1 + \cot^2 A = \operatorname{cosec}^2 A$
10	Values of sin (0°, 30°, 45°, 60°, 90°)	$0,\ \frac{1}{2},\ \frac{1}{\sqrt{2}},\ \frac{\sqrt{3}}{2},\ 1$
11	Values of cos (0°, 30°, 45°, 60°, 90°)	$1,\ \frac{\sqrt{3}}{2},\ \frac{1}{\sqrt{2}},\ \frac{1}{2},\ 0$
12	Values of tan (0°, 30°, 45°, 60°, 90°)	$0,\ \frac{1}{\sqrt{3}},\ 1,\ \sqrt{3},\ \text{not defined}$
13	Range of sin and cos	$0 \le \sin A \le 1$ and $0 \le \cos A \le 1$ for $0^\circ \le A \le 90^\circ$
14	Useful factor forms	$(\sec A + \tan A)(\sec A - \tan A) = 1$; $(\operatorname{cosec} A + \cot A)(\operatorname{cosec} A - \cot A) = 1$